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4.4 Answer Explanations

The following discussion is intended to familiarize you with the most efficient and effective approaches to the kinds of problems common to Quantitative Reasoning questions. The particular questions in this chapter are generally representative of the kinds of Quantitative Reasoning questions you will encounter on the GMAT exam. Remember that it is the problem-solving strategy that is important, not the specific details of a particular question.

Questions 1 to 96 — Difficulty: Easy

1. In the graduating class of a certain college, 48 percent of the students identify exclusively as male and 52 percent identify exclusively as female. In this class, 40 percent of the students who identify as male and 20 percent of the students who identify as female are 25 years old or older. If one student in the graduating class is randomly selected, approximately what is the probability that the student will be less than 25 years old?

   A. 0.90

   B. 0.70

   C. 0.45

   D. 0.30

   E. 0.25

   Arithmetic Statistics

   In the class, the percentage of students who identify exclusively as male and are less than 25 years old is 48% × (100 − 40)% = 28.8%. The percentage who identify exclusively as female and are less than 25 years old is 52% × (100 − 20)% = 41.6%. So, the percentage of students who are less than 25 years old is 28.8% + 41.6% = 70.4% ≈ 70%. Thus, the probability of a randomly selected student in the class being less than 25 years old is about 70%.

   The correct answer is B.

2. In a certain board game, a stack of 48 cards, 8 of which represent a single share of stock, are shuffled and then placed face down. If the first 2 cards selected do not represent shares of stock, what is the probability that the third card selected will represent a share of stock?

   A. $\frac{1}{8}$

   B. $\frac{1}{6}$

   C. $\frac{1}{5}$

   D. $\frac{3}{23}$

   E. $\frac{4}{23}$

   Arithmetic Statistics

   After the first 2 cards have been drawn, 46 cards remain, including all 8 of those representing one share of stock apiece. So, assuming that any of those 46 cards is equally likely to be the third card drawn, the probability that the third card drawn represents a share of stock is $\frac{8}{46} = \frac{4}{23}$.

   The correct answer is E.

3. During a trip that they took together, Carmen, Juan, Maria, and Rafael drove an average (arithmetic mean) of 80 miles each. Carmen drove 72 miles, Juan drove 78 miles, and Maria drove 83 miles. How many miles did Rafael drive?

   A. 80

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A. 80

B. 82

C. 85

D. 87

E. 89

Arithmetic Statistics

Let C, J, M, and R be the numbers of miles, respectively, that Carmen, Juan, Maria, and Rafael drove. Since the average of the numbers of miles they drove is 80, it follows that $\frac{C+J+M+R}{4} = 80$, or $C+J+M+R = 4(80) = 320$. It is given that $C = 72$, $J = 78$, and $M = 83$. Therefore, $72 + 78 + 83 + R = 320$, or $R = 87$.

The correct answer is D.

4. Each week, a clothing salesperson receives a commission equal to 15 percent of the first $500 in sales and 20 percent of all additional sales that week. What commission would the salesperson receive on total sales for the week of $1,300?

A. $195

B. $227

C. $235

D. $260

E. $335

Arithmetic Applied Problems

The commission on the total sales can be calculated as follows:

commission on first $500 + commission on amount over $500

= (0.15)($500) + (0.20)($1,300 − $500)

= $75 + $160

Therefore, the commission on the total sales is $75 + $160 = $235.

The correct answer is C.

5. Five batches of 100 nails each are taken from a production line. The numbers of defective nails in the first four batches are 2, 4, 3, and 5, respectively. If the fifth batch has either 1, 2, or 6 defective nails, for which of these values does the average (arithmetic mean) number of defective nails per batch for the five batches equal the median number of defective nails for the five batches?

I. 1

II. 2

III. 6

A. I only

B. II only

C. III only

D. I and III only

E. I, II, and III

Arithmetic Statistics

If the fifth batch has just 1 defective nail, the numbers of defective nails in the five batches are 2, 4, 3, 5, and 1. Then the mean number of defective nails per batch is $\frac{(1+2+3+4+5)}{5} = \frac{15}{5} = 3$. The median is also 3, since exactly 2 batches have fewer than 3 defective nails apiece and exactly 2 have more than 3 defective nails

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apiece. So, in this first case, the median and mean are equal. Now consider the second case, in which the fifth batch has 2 defective nails. In this case, the mean number of defective nails per batch is $\frac{16}{5} = 3.2$, whereas the median is still 3. In this second case, the median and the mean are not equal. And in the third case, the fifth batch has 6 defective nails, so the mean number of defective nails per batch is $\frac{20}{5} = 4$. The median is also 4, since 2 batches have fewer than 4 defective nails apiece and 2 have more than 4 apiece. Thus, in the third case, the median and mean are equal.

The correct answer is D.

6. List S consists of 10 consecutive odd integers, and list T consists of 5 consecutive even integers. If the least integer in S is 7 more than the least integer in T, how much greater is the average (arithmetic mean) of the integers in S than the average of the integers in T?

   A. 2

   B. 7

   C. 8

   D. 12

   E. 22

Arithmetic Statistics

Let the integers in S be $s, s+2, s+4, \ldots, s+18$, where s is odd. Let the integers in T be $t, t+2, t+4, t+6, t+8$, where t is even. Given that $s = t + 7$, it follows that $s - t = 7$. The average of the integers in S is $\frac{10s+90}{10} = s + 9$, and, similarly, the average of the integers in T is $\frac{5t+20}{5} = t + 4$. The difference in these averages is $(s+9) - (t+4) = (s-t) + (9-4) = 7 + 5 = 12$. Thus, the average of the integers in S is 12 greater than the average of the integers in T.

The correct answer is D.

Lane number	Number in line
1	5
2	4
3	3
4	4
5	2

7. A certain grocery store has 5 checkout lanes. The table above shows the number of people who were waiting in line at each checkout lane at closing time yesterday. What was the median number of people waiting in line?

   A. $2\frac{4}{5}$

   B. 3

   C. $3\frac{1}{2}$

   D. $3\frac{3}{5}$

   E. 4

Arithmetic Statistics

To find the median number of people in line in the 5 checkout lanes, we take the 5 numbers of people in line, shown in the table's "Number in line" column, and list them from least to greatest: 2, 3, 4, 4, 5. The middle number in this list is a 4, so 4 is the median.

The correct answer is E.

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The correct answer is E.

8. $\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+\left(\frac{1}{4}-\frac{1}{5}\right)+\left(\frac{1}{5}-\frac{1}{6}\right)=$

A. $-\frac{1}{6}$

B. 0

C. $\frac{1}{3}$

D. $\frac{1}{2}$

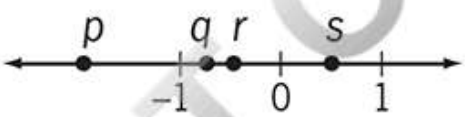
E. $\frac{2}{3}$

Arithmetic Operations with Rational Numbers

The parentheses can be removed without any change of signs, and after doing this most of the terms can be additively cancelled as shown below.

$$\frac{1}{2}-\cancel{\frac{1}{3}}+\cancel{\frac{1}{3}}-\cancel{\frac{1}{4}}+\cancel{\frac{1}{4}}-\cancel{\frac{1}{5}}+\cancel{\frac{1}{5}}-\frac{1}{6}=\frac{1}{2}-\frac{1}{6}=\frac{1}{3}$$

The correct answer is C.



9. If p, q, r, and s are as shown on the number line above, which of the following products is greatest?

A. pq

B. pr

C. qr

D. qs

E. rs

Arithmetic Number Line

The number line shows that p, q, and r are all negative, while s is positive. The product of any two negative numbers is positive. That means pq, pr, and qr are each positive, so each is greater than qs and rs, which are both negative. And since $p < q < r < 0$, then $pq > pr > qr$. Thus, pq is the greatest of the 5 products.

The correct answer is A.

10. Last year $48,000 of a certain store's profit was shared by its 2 owners and their 10 employees. Each of the 2 owners received 3 times as much as each of their 10 employees. How much did each owner receive from the $48,000?

A. $12,000

B. $9,000

C. $6,000

D. $4,000

E. $3,000

Algebra First-Degree Equations

Let A be the amount received by each owner and let B be the amount received by each employee. From the given information, it follows that $A = 3B$ and $2A + 10B = 48,000$. Thus, $2(3B) + 10B = 48,000$, or

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given information, it follows that $A = 3B$ and $2A + 10B = 48{,}000$. Thus, $2(3B) + 10B = 48{,}000$, or $16B = 48{,}000$, or $B = 3{,}000$. Therefore, the amount received by each owner was $A = 3B = 3(\$3{,}000) = \$9{,}000$.

The correct answer is B.

11. On a vacation, Rose exchanged $500.00 for euros at an exchange rate of 0.80 euro per dollar and spent $\frac{3}{4}$ of the euros she received. If she exchanged the remaining euros for dollars at an exchange rate of $1.20 per euro, what was the dollar amount she received?

A. $60.00

B. $80.00

C. $100.00

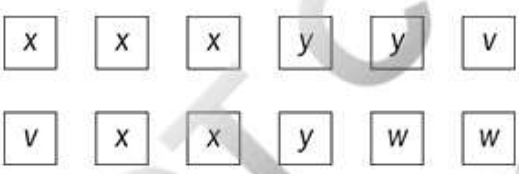
D. $120.00

E. $140.00

Arithmetic Operations with Rational Numbers

At the exchange rate of 0.80 euro per dollar, Rose exchanged $500.00 for (0.80)(500) = 400 euros. She spent $\frac{3}{4}\left(400\right) = 300$ euros, had 400 – 300 = 100 euros left, and exchanged them for dollars at the exchange rate of $1.20 per euro. Therefore, the dollar amount she received was (1.20)(100) = $120.00.

The correct answer is D.



12. Each of the 12 squares shown is labeled x, y, v, or w. What is the ratio of the number of these squares labeled x or y to the number of these squares labeled v or w?

A. 1:2

B. 2:3

C. 4:3

D. 3:2

E. 2:1

Arithmetic Ratio and Proportion

By a direct count, there are 8 squares labeled x or y (5 labeled x, 3 labeled y) and there are 4 squares labeled v or ω (2 labeled v, 2 labeled ω). Therefore, the ratio of the number of squares labeled x or y to the number of squares labeled v or ω is 8:4, which reduces to 2:1.

The correct answer is E.

Number of Solid-Colored Marbles in Three Jars			
Jar	Number of red marbles	Number of green marbles	Total number of red and green marbles
P	x	y	80
Q	y	z	120
R	x	z	160

13. In the table above, what is the number of green marbles in Jar R?

A. 70

B. 80

C. 90

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C. 90

D. 100

E. 110

Arithmetic; Algebra Interpretation of Tables; Applied Problems

First, set up an equation to find the total number of marbles in the three jars as follows:

$$x + y + y + z + x + z = 80 + 120 + 160$$

$$2x + 2y + 2z = 360 \text{ combine the like terms}$$

$$x + y + z = 180 \text{ divide both sides by 2}$$

Then, since it can be seen from the table that the number of green marbles in Jar R is z, solve for z to answer the problem. To do this most efficiently, use the information from the table for Jar P, which is that $x + y = 80$.

$$x + y + z = 180$$

$$80 + z = 180 \text{ substitute 80 for } x + y$$

$$z = 100$$

The correct answer is D.

14. Bouquets are to be made using white tulips and red tulips, and the ratio of the number of white tulips to the number of red tulips is to be the same in each bouquet. If there are 15 white tulips and 85 red tulips available for the bouquets, what is the greatest number of bouquets that can be made using all the tulips available?

A. 3

B. 5

C. 8

D. 10

E. 13

Arithmetic Applied Problems; Properties of Numbers

Because all the tulips are to be used and the same number of white tulips will be in each bouquet, the number of white tulips in each bouquet times the number of bouquets must equal the total number of white tulips, or 15. Thus, the number of bouquets must be a factor of 15, and so the number must be 1, 3, 5, or 15. Also, the number of red tulips in each bouquet times the number of bouquets must equal the total number of red tulips, or 85. Thus, the number of bouquets must be a factor of 85, and so the number must be 1, 5, 17, or 85. Since the number of bouquets must be 1, 3, 5, or 15, and the number of bouquets must be 1, 5, 17, or 85, it follows that the number of bouquets must be 1 or 5, and thus the greatest number of bouquets that can be made is 5. Note that each of the 5 bouquets will have 3 white tulips, because (5)(3) = 15, and each of the 5 bouquets will have 17 red tulips, because (5)(17) = 85.

The correct answer is B.

County	Amount Recycled	Amount Disposed of
A	16,700	142,800
B	8,800	48,000
C	13,000	51,400
D	3,900	20,300
E	3,300	16,200

15. The table above shows the amount of waste material, in tons, recycled by each of five counties in a single year and the amount of waste material, also in tons, that was disposed of in landfills by the five counties in that year. Which county had the lowest ratio of waste material disposed of to waste material recycled in the year reported in the table?

A. A

B. B

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B. B

C. C

D. D

E. E

Arithmetic Ratio and Proportion

To quickly estimate each county's ratio of waste material disposed of to waste material recycled, find the row representing that county in the table. Then in that row, roughly estimate the ratio of the number in the "Amount Disposed of" column to that in the "Amount Recycled" column. You can just divide by 1,000 and round to get nice simple ratios. County A's ratio (142,800:16,700) is more than 140:20, which is 7:1. County B's is more than 45:9, which is 5:1. County C's is less than 52:13, which is 4:1. County D's is more than 20:4, which is 5:1. County E's is more than 16:4, which is 4:1. Thus, County C has the lowest ratio of material disposed of to material recycled.

The correct answer is C.

16. 125% of 5 =

A. 5.125

B. 5.25

C. 6

D. 6.125

E. 6.25

Arithmetic Percents

125% of 5 represents $\frac{125}{100} \times 5$, or $1.25 \times 5 = 6.25$.

The correct answer is E.

17. Each day Linda spends 6 hours at her dog-walking job, for which she is paid $15.00 per hour. In addition, Linda spends a total of 1.5 hours each day, (for which she is not paid), traveling to and from the job. What is Linda's hourly rate of pay for the total amount of time that she spends each day at her dog-walking job and traveling to and from that job?

A. $7.50

B. $9.00

C. $10.00

D. $10.75

E. $12.00

Arithmetic Applied Problems

Linda is paid $15/hour for 6 hours of dog-walking job each day, totaling $90. She's not paid for her 1.5 hours of commuting each day. So, her total daily pay for the dog-walking job and commuting combined is $90 for 7.5 hours. Thus, her hourly rate of pay for the dog-walking job and commuting combined is $90/7.5 hours = $12/hour.

The correct answer is E.

18. When traveling at a constant speed of 32 miles per hour, a certain motorboat consumes 24 gallons of fuel per hour. What is the fuel consumption of this boat at this speed measured in miles traveled per gallon of fuel?

A. $\frac{2}{3}$

B. $\frac{3}{4}$

C. $\frac{4}{5}$

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C. $\frac{}{5}$

D. $\frac{4}{3}$

E. $\frac{3}{2}$

Arithmetic Operations with Rational Numbers

If the motorboat consumes 24 gallons of fuel in 1 hour, then it consumes 1 gallon of fuel in $\frac{1}{24}$ hour. If the motorboat travels 32 miles in 1 hour, then it travels $\frac{32}{24} = \frac{4}{3}$ miles in $\frac{1}{24}$ hour, which is the length of time it takes to consume 1 gallon of fuel. Thus, the motorboat travels $\frac{4}{3}$ miles per gallon of fuel.

The correct answer is D.

19. If snow accumulation increased at a constant rate of 30 millimeters per hour during a certain snowstorm, how many seconds did it take for snow accumulation to increase by 1 millimeter?

A. $\frac{1}{120}$

B. $\frac{1}{60}$

C. $\frac{1}{20}$

D. 20

E. 120

Arithmetic Applied Problems

Since the snow accumulation increased at a constant 30 mm/hour, a 1-mm increase in snow accumulation took 1/30 of an hour, which is 2 minutes or 120 seconds.

The correct answer is E.

20. From 2000 to 2003, the number of employees at a certain company increased by a factor of $\frac{1}{4}$. From 2003 to 2006, the number of employees at this company decreased by a factor of $\frac{1}{3}$. If there were 100 employees at the company in 2006, how many employees were there at the company in 2000?

A. 200

B. 120

C. 100

D. 75

E. 60

Algebra First-Degree Equations

Let N be the number of employees in 2000. In 2003 there were $\frac{1}{4}$ (number in 2000) = $\frac{1}{4}N$ more employees than in 2000, for a total of $N + \frac{1}{4}N = \frac{5}{4}N$ employees. In 2006 there were $\frac{1}{3}$ (number in 2003) = $\frac{1}{3}\left(\frac{5}{4}\right)N$ fewer employees than in 2003, for a total of $\frac{5}{4}N - \frac{1}{3}\left(\frac{5}{4}N\right) = \frac{5}{6}N$ employees. It is given that there were 100 employees in 2006, so $\frac{5}{6}N = 100$, or $N = \frac{6}{5}(100) = 120$.

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6 5

The correct answer is B.

21. Working alone at its own constant rate, machine R fills 10,000 boxes with pins in 9 hours, and working alone at its own constant rate, machine S fills 5,000 boxes with pins in 3 hours. If machine R and machine S, each working at its own constant rate and for the same period of time, together filled a certain number of boxes with pins, what percent of the boxes were filled by machine R?

    A. 25%

    B. 33%

    C. 40%

    D. 66%

    E. 75%

    Arithmetic Applied Problems

    In the 9 hours machine R takes to fill 10,000 boxes of pins, machine S fills (5,000 boxes/3 hours) × 3 hours = 15,000 boxes of pins. So, each working at its own constant rate for 9 hours, the two machines together fill 10,000 + 15,000 = 25,000 boxes of pins. Therefore, machine R fills 10,000/25,000 = 0.4 = 40% of that total.

    The correct answer is C.

22. A collection of 16 coins, each with a face value of either 10 cents or 25 cents, has a total face value of $2.35. How many of the coins have a face value of 25 cents?

    A. 3

    B. 5

    C. 7

    D. 9

    E. 11

    Algebra First-Degree Equations

    Let x represent the number of coins each with a face value of 25 cents. Then, since there are 16 coins in all, $16-x$ represents the number of coins each with a face value of 10 cents. The total face value of the coins is $2.35 or 235 cents so,

$$\begin{aligned} 25x + 10(16-x) &= 235 && \text{given} \\ 25x + 160 - 10x &= 235 && \text{distributive property} \\ 15x + 160 &= 235 && \text{combine like terms} \\ 15x &= 75 && \text{subtract 160 from both sides} \\ x &= 5 && \text{divide both sides by 15} \end{aligned}$$

    Therefore, 5 of the coins have a face value of 25 cents.

    The correct answer is B.

23. A retailer purchased eggs at $2.80 per dozen and sold the eggs at 3 eggs for $0.90. What was the retailer's gross profit from purchasing and selling 5 dozen eggs? (1 dozen eggs = 12 eggs)

    A. $0.90

    B. $2.40

    C. $4.00

    D. $11.30

    E. $12.00

    Arithmetic Applied Problems

    The retailer's cost was $2.80 per dozen eggs and the retailer's revenue was $0.90 per 3 eggs, or 4($0.90) = $3.60 per dozen eggs. Therefore, the retailer's profit for 5 dozen eggs—revenue minus cost for 5 dozen eggs—was

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per dozen eggs. Therefore, the retailer's profit for 5 dozen eggs—revenue minus cost for 5 dozen eggs—was 5($3.60 – $2.80) = 5($0.80) = $4.00.

The correct answer is C.

24. In a set of 24 cards, each card is numbered with a different positive integer from 1 to 24. One card will be drawn at random from the set. What is the probability that the card drawn will have either a number that is divisible by both 2 and 3 or a number that is divisible by 7?

A. $\frac{3}{24}$

B. $\frac{4}{24}$

C. $\frac{7}{24}$

D. $\frac{8}{24}$

E. $\frac{17}{24}$

Arithmetic Probability

The desired probability is N divided by 24, where N is the number of positive integers from 1 through 24 that are either divisible by both 2 and 3,or divisible by 7. Since an integer is divisible by both 2 and 3 if and only if the integer is divisible by 6, it follows that N is the number of positive integers from 1 through 24 that are either divisible by 6 or divisible by 7. There are 4 numbers from 1 through 24 that are divisible by 6, namely 6, 12, 18, and 24. There are 3 numbers from 1 through 24 that are divisible by 7, namely 7, 14, and 21. Since these numbers are all different from each other, it follows that $N = 4 + 3 = 7$ and the desired probability is $\frac{N}{24} = \frac{7}{24}$.

The correct answer is C.

25. As a salesperson, Phyllis can choose one of two methods of annual payment: either an annual salary of $35,000 with no commission or an annual salary of $10,000 plus a 20% commission on her total annual sales. What must her total annual sales be to give her the same annual pay with either method?

A. $100,000

B. $120,000

C. $125,000

D. $130,000

E. $132,000

Algebra Applied Problems

Letting s be Phyllis's total annual sales needed to generate the same annual pay with either method, the given information can be expressed as $\$35{,}000 = \$10{,}000 + 0.2s$. Solve this equation for s.

$$\$35{,}000 = \$10{,}000 + 0.2s$$
$$\$25{,}000 = 0.2s$$
$$\$125{,}000 = s$$

The correct answer is C.

26. If $1 < x < y < z$, which of the following has the greatest value?

A. $z(x+1)$

B. $z(y+1)$

C. $x(y+z)$

D. [illegible]

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D. $y(x+z)$

E. $z(x+y)$

Algebra Inequalities

This problem can be solved by calculating each of the options for a fixed and appropriate choice of values of the variables. For example, if $x=2$, $y=3$, and $z=4$, then $1<x<y<z$ and the values of the options are as follows:

A. $z(x+1)=4(2+1)=12$

B. $z(y+1)=4(3+1)=16$

C. $x(y+z)=2(3+4)=14$

D. $y(x+z)=3(2+4)=18$

E. $z(x+y)=4(2+3)=20$

Alternatively, this problem can also be solved algebraically.

Answer choice E is greater than each of answer choice A and B:

$$z(x+1)=zx+z \quad < \quad zx+zy=z(x+y)$$

Answer choice A use $1<y$ Answer choice E

$$z(y+1)=zy+z \quad < \quad zy+zx=z(y+1)$$

Answer choice B use $1<x$ Answer choice E

Answer choice E is greater than each of answer choice C and D:

$$x(y+z)=xy+xz \quad < \quad zy+xz=z(y+x)$$

Answer choice C use $x<z$ Answer choice E

$$y(x+z)=yx+yz \quad < \quad zx+yz=z(x+y)$$

Answer choice D use $y<z$ Answer choice E

The correct answer is E.

27. Set X consists of eight consecutive integers. Set Y consists of all the integers that result from adding 4 to each of the integers in Set X and all the integers that result from subtracting 4 from each of the integers in Set X. How many more integers are there in Set Y than in Set X?

A. 0

B. 4

C. 8

D. 12

E. 16

Arithmetic Operations with Integers

Since the 8 consecutive integers in Set X are not specified, let Set $X=\{3,4,5,6,7,8,9,10\}$. The following table shows Set X and Set Y whose elements were obtained by adding 4 to each element of Set X and subtracting 4 from each element of Set X. The elements of Set Y that are also in Set X are marked with an asterisk, *. There are 8 unmarked elements in Set Y, so Set Y has 8 more elements than Set X.

Set X	Set Y	
	Add 4	Subtract 4
3	7*	–1
4	8*	0
5	9*	1

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5	9*	1
6	10*	2
7	11	3*
8	12	4*
9	13	5*
10	14	6*

The correct answer is C.

28. Of the following, which is the closest to $\frac{60.2}{1.03 \times 4.86}$?

A. 10

B. 12

C. 13

D. 14

E. 15

Arithmetic Estimation

Replace the three numbers appearing in the expression with three nearby integers that allow the arithmetic operations to be carried out easily to get an approximation. $\frac{60.2}{1.03 \times 4.86} \approx \frac{\cancel{60}^{12}}{1 \times \cancel{5}^{1}} = \frac{12}{1 \times 1} = 12$

The correct answer is B.

29. Thabo owns exactly 140 books, and each book is either paperback fiction, paperback nonfiction, or hardcover nonfiction. If he owns 20 more paperback nonfiction books than hardcover nonfiction books, and twice as many paperback fiction books as paperback nonfiction books, how many hardcover nonfiction books does Thabo own?

A. 10

B. 20

C. 30

D. 40

E. 50

Algebra Simultaneous First-Degree Equations

Let F represent the number of paperback fiction books that Thabo owns; N_p, the number of paperback nonfiction books; and N_h, the number of hardcover nonfiction books. It is given that $F + N_p + N_h = 140$, $N_p = N_h + 20$, and $F = 2N_p = 2(N_h + 20)$. It follows that

$$\begin{aligned} F + N_p + N_h &= 140 \text{ given} \\ 2(N_h + 20) + (N_h + 20) + N_h &= 140 \text{ by substitution} \\ 4N_h + 60 &= 140 \text{ combine like terms} \\ 4N_h &= 80 \text{ subtract 60 from both sides} \\ N_h &= 20 \text{ divide both sides by 4} \end{aligned}$$

The correct answer is B.

30. What is 35 percent of the sum of 1.4 and $\frac{1}{5}$?

A. 0.42

B. 0.56

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B. 0.56

C. 0.85

D. 1.55

E. 1.95

Arithmetic Applied Problems

Thirty-five percent of the sum of 1.4 and $\frac{1}{5}$ is 0.35(1.4 + 0.2) = 0.35(1.6) = 0.56.

The correct answer is B.

31. If $x > 0$, $\frac{x}{50} + \frac{x}{25}$ is what percent of x?

A. 6%

B. 25%

C. 37%

D. 60%

E. 75%

Algebra Arithmetic; Simplifying Algebraic Expressions; Percents

Because we want a percent, use a common denominator of 100 to combine the two terms.

$\frac{x}{50} + \frac{x}{25} = \frac{2x}{100} + \frac{4x}{100} = \frac{6x}{100} = \left(\frac{6}{100}\right)x$, which is 6% of x.

The correct answer is A.

32. During a certain time period, Car X traveled north along a straight road at a constant rate of 1 mile per minute and used fuel at a constant rate of 5 gallons every 2 hours. During this time period, if Car X used exactly 3.75 gallons of fuel, how many miles did Car X travel?

A. 36

B. 37.5

C. 40

D. 80

E. 90

Arithmetic Applied Problems

The car used fuel at the rate of 5 gallons every 2 hours, and thus after using 3.75 gallons of fuel, the car had traveled for $\left(3.75\right)\left(\frac{2}{5}\right)$ hours. One way to obtain this value is indicated by the calculations in the following table.

5 gal	corresponds to	2 hr	Given
1 gal	corresponds to	$\frac{2}{5}$hr	Divide by 5
3.75 gal	corresponds to	$\left(3.75\right)\left(\frac{2}{5}\right)$ hr	Multiply by 3.75

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